

Near-Lossless Long Context under Fixed VRAM via Critical-Span Retention

Workshop / arXiv-style draft

Lab: RTX 3090 24 GB Primary: Qwen/Qwen3-4B-Instruct-2507

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Abstract

Long-context inference is limited by KV-cache memory. Training-free eviction methods shrink the cache, but it is often unclear *which* tokens are necessary for near-full-KV quality, and whether online streaming fails from insufficient peak budget or from weak **query-unknown discovery**. On a multi-seed controlled retrieval suite (Qwen3-4B, RTX 3090), we show:

1. **Structure (H1')**. Retaining critical fact tokens plus a small local radius $R^* = 1$ (with attention sinks and a recent/question window) is **necessary and sufficient** for $\varepsilon \approx 0$ single-fact recall: oracle **15/15**, anti-oracle **0/15**, full KV **15/15** (5 seeds $\times$ 3 depths @4k).
2. **Discovery gap**. Online streaming at moderate peak fails under multi-seed hay not because the budget is too small, but because discovery is weak: attention stream@512 is **33%**, while perfect online pin of critical $\pm R$ is **15/15** at the same peak.
3. **Closing the gap (suite)**. A **surface novelty detector** (within-prefix rarity + digit/ID-like cues; no final question) with **sticky pin packing** reaches $\sim$ **93%** multi-seed @4k and **100%** (3×3) through **40k** at stream@512—peak cache $\sim$ **1k tokens**, flat in L .
4. **Public long-doc QA (decomposition)**. On a 60-item LongBench slice, sticky novelty lags full ($\sim 0.64 \times$ F1 at peak $\sim 1k$). A **posthoc query-aware upper bound** (full prefill $\rightarrow$ compress) recovers **0.95–1.0** $\times$ full F1 at final $B \in \{512, 1024\}$, proving discovery *can* work when the question is known. **Query-hold** streaming trades peak for quality, best $\approx$ **0.92** $\times$ full F1 at peak $\sim 2.5k$.

The critical-span mechanism transfers across Qwen2.5, Llama-3.2, and hybrid Gemma-4 (full layers). We do **not** claim general long-context SOTA; see Section 6.

1 Introduction

KV cache memory grows with context length L . On a fixed workstation (here: 24 GB), full-cache decode becomes the bottleneck long before “the model cannot attend.” Training-free compressors reduce tokens kept, but practice often optimizes average scores rather than **guaranteeing** the spans that encode answers—and online pipelines must score tokens **before** the user question exists.

We study **near-lossless** compression on retrieval-critical prompts: quality Q should match full KV at $\varepsilon \approx 0$ under a multi-seed protocol. The systems objective is

$$L_\varepsilon = \max \{ L : Q(M, L) \geq (1 - \varepsilon) Q(\text{Full}, L) \} \quad (1)$$

under **peak** cache / VRAM constraints—not only final decode size after a full prefill.

Claim. Near-lossless training-free KV compression is better understood as **retaining critical local neighborhoods** and **discovering them online before the question exists** than as uniform thinning or posthoc score ranking alone. On open long-document QA, the residual gap is largely an **online retention / peak-cache Pareto**, not a failure of query-aware scoring once the question is available.

2 Related work

Training-free KV eviction and selection. A large line of work compresses the KV cache without fine-tuning by dropping low-scoring tokens. **H2O** [1] keeps heavy hitters under accumulated attention; **StreamingLLM** [2] observes attention sinks and retains initial tokens plus a sliding recent window; **Scissorhands** [3] and **TOVA** [4] prune by recency/attention proxies; **SnapKV** [5] and pyramid-style keepers [6] select important prefix tokens using observation-window attention, often with clustering or hierarchical budgets. These methods primarily address *which scores to use when queries (or an observation window) exist*. Our contribution is complementary: multi-seed **kill tests** that isolate *necessary keep sets*, an explicit **query-unknown discovery gap** for online streaming, and a lightweight surface detector + sticky packing that closes that gap on retrieval suites at flat peak cache.

Streaming and long-context systems. Chunked prefill and stream compression cap **peak** memory rather than only final decode size. We report peak cache tokens, peak VRAM, and prefill latency on a consumer GPU, in the spirit of systems-oriented long-context work rather than leaderboard-only evaluation. Hybrid architectures (sliding-window + full-attention layers; e.g. Gemma-4 class models) still leave full-layer KV as the dominant compressible state; we compress full layers only and keep bookkeeping for hybrid score passes.

Quantization. KV quantization (e.g. **KIVI** [7]) reduces bytes per retained token and is complementary to eviction. We use fake-int8 only for logical byte accounting in some benches; quality results use bf16 caches unless noted.

Benchmarks. Needle-in-a-haystack and multi-hop synthetic suites stress retrieval under controlled depth/seed. **LongBench** [8] and related public suites (RULER, InfiniteBench) stress general long-document and multi-doc QA. We use a **fixed 60-item LongBench slice** (not a full leaderboard) to separate suite success from open QA, and to measure posthoc upper bounds and peak/quality Pareto curves.

Positioning. We are not proposing a new attention kernel or claiming SnapKV-beating LongBench SOTA. The paper’s contribution is a **mechanism + discovery diagnosis + workstation L_ϵ operating point**, with honest public-data limits and a quantitative online Pareto.

3 Methods (short)

Task (suite). Needle-in-haystack style secrets in seeded filler; depths $\{0, 0.5, 1\}$; success = exact key recall (greedy decode). Stresses: NL facts, multi-needle, multi-hop, prose soft facts, multidoc.

H1’ keep set. $\text{sinks} \cup \text{critical} \pm R \cup \text{recent/question window}$. Kill arms: full, oracle_r1, anti_oracle.

Posthoc scorer. `seed_valley`: shared obs-window attention $\rightarrow$ seeds $\rightarrow$ contiguous valley + $\pm R$. Used both for suite tax tables and as the LongBench **posthoc upper bound** after full prefill.

Streaming. Chunked prefill; compress when cache $>$ budget; peak $\approx$ budget + chunk.

Oracle-online upper bound. Force-keep $\text{critical} \pm R$ whenever still in cache (perfect discovery).

Surface novelty (query-unknown). Per-token score from within-prefix rarity, first occurrence, and surface cues (digits, ID-like pieces). Expand peaks $\pm R$ into pin sets. **Sticky** registry: once discovered, pins stay until evicted; packing ranks pins by score under budget (sinks + recent reserved).

Near-lossless KV: critical spans → discovery gap → sticky novelty

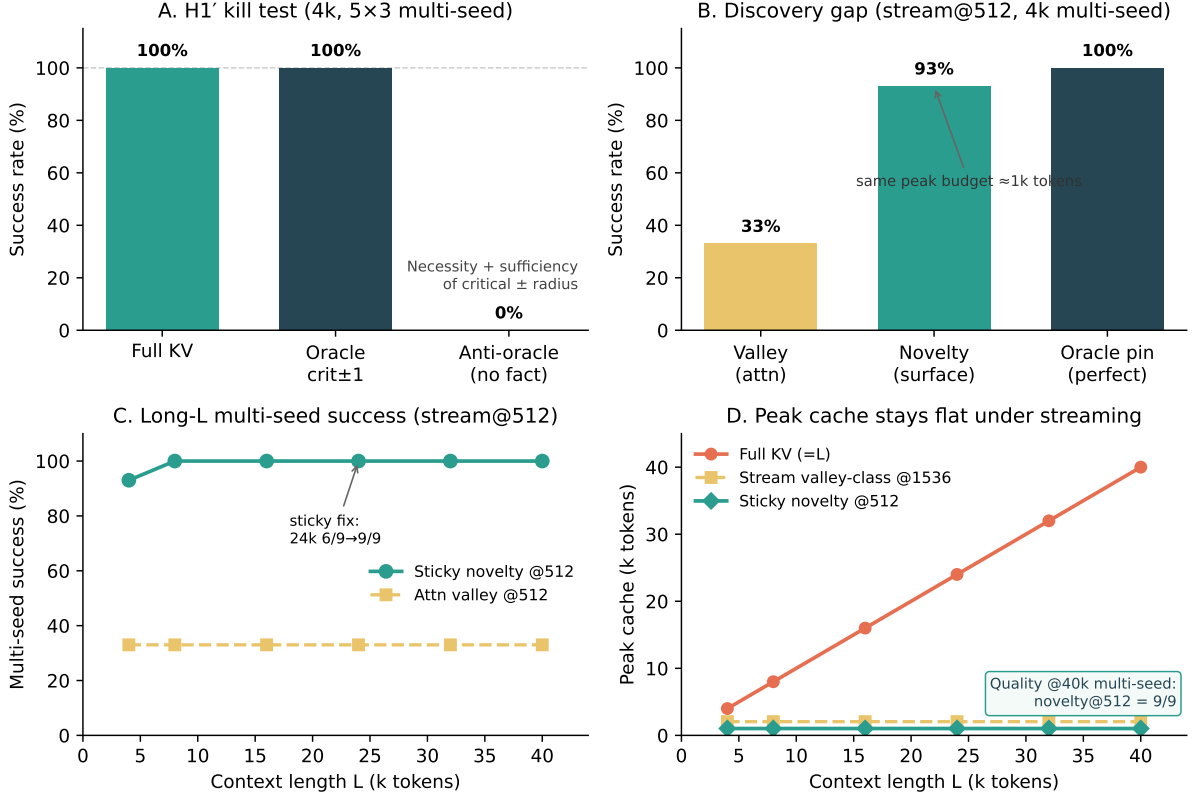


Figure 1: **Story figure.** (A) H1' kill test @4k multi-seed (5×3): full and oracle crit±1 both 15/15; anti-oracle 0/15. (B) Discovery gap at stream@512: valley 33%, surface novelty 93%, oracle pin 100%. (C) Long- L multi-seed success: sticky novelty@512 holds 9/9 through 40k; valley@512 stays end-only (~33%). (D) Peak cache tokens stay flat (~1k) under sticky novelty stream@512 while full KV grows with L .

Query-hold (systems lever). Mid-stream hybrid pins under a larger `hold_budget`, then query-aware tighten to `final_budget` once the question is in the observation window. Peak $\approx$ hold + chunk; quality trades against peak.

Public LongBench slice. THUDM/LongBench: 20 items each of `multifieldqa_en`, `qasper`, `hotpotqa` (60 total), contexts truncated to max 4096 tokens; token F1 vs gold + substring hit; greedy decode, `max_new=64`.

Default API. `prefill_auto(..., mode="stream", discovery="novelty").`
Use `discovery="query_hold"` for the peak/quality Pareto path.

4 Results

4.1 Figure 1 — Story in four panels

A. H1' kill (4k, 5×3 multi-seed). Full and oracle crit±1 both **15/15**; anti-oracle **0/15**. Bare critical tokens without radius often corrupt trailing digits (H1 needs local context); $R^* = 1$ restores $\varepsilon \approx 0$. Mean $|\text{oracle}| \approx 149$ tokens vs full $\sim 4k$.

B. Discovery gap (stream@512). Attention valley **33%**; surface novelty **93%** (14/15); oracle pin **100%**. Peak budget is in the same class ($\sim 1k$ with chunk). Failures of valley at 512 are **detector** failures, not “stream cannot work.”

C. Long L multi-seed (3×3). Sticky novelty@512 is **9/9** at 16k, 24k, 32k, and **40k**. Non-sticky novelty degraded at 24k ($\sim 6/9$) via re-rank thrash; sticky packing fixed it. Valley@512 remains end-depth-only ($\sim 33\%$) under multi-seed hay as L grows.

D. Peak cache. Full KV scales with L ; sticky novelty stream@512 stays $\sim 1k$ tokens. Prior valley operating points often used $\sim 1.5\text{--}2k$ peak for long L .

4.2 Scorer tax (posthoc, question known)

Table 1: Posthoc budget tax vs oracle keep set (4k multi-seed).

Method	All-cell $B_{\min}$	Tax vs mean oracle
oracle_r1	~ 149	$1.0\times$
seed_valley	192	$\sim 1.24\times$ global / $\sim 1.16\times$ mean

Posthoc is near-oracle-tight; **stream is not**, until discovery improves.

4.3 Stress and transfer (summary)

Table 2: Stress suite and cross-model transfer (summary).

Check	Result
NL facts (names/places) sticky novelty@512	15/15 multi-seed
Code needles sticky	15/15
Adversarial ID-like filler (pre-sticky)	novelty 100% vs valley 33%
Multi-needle recall_all @512	novelty 5/5 ($\max_{\text{new}} \geq 96$); valley 0/5
2-hop (Alice $\rightarrow$ id $\rightarrow$ password) @512	novelty 5/5 ; valley 1/5 (needs @1024 for 5/5)
3-hop + distractors @512	novelty 5/5 ; valley 0/5 (needs @1024 for 5/5)
Prose soft fact (no digits) @512	novelty 15/15 ; valley 53%
Multidoc (6 titled docs) @512	novelty 15/15 ; valley/oracle_pin 5/15
External-style mixed QA ($10 \times \sim 4k$)	novelty 10/10 hits (= full); valley 0/10
Gemma-4 E4B hybrid novelty@512	multi-seed 9/9 @4k (valley needs ~ 1024)
Qwen2.5 / Llama-3.2	H1 holds; family-specific posthoc floors

4.4 Public LongBench decomposition

Synthetic suite success does not automatically transfer to open long-document QA. On the fixed 60-item slice (same items for all arms), sticky novelty stream@512 is only slightly above attention streaming and far below full (Table 3). A posthoc query-aware upper bound—full prefill, then `seed_valley` compress with the question in the observation window—recovers **$0.95\times$** full mean F1 at final $B=512$ and $\sim 1.0\times$ at 1024/2048. Peak remains full L for posthoc; the point is diagnostic: **query-aware scoring can keep answer-bearing tokens** when allowed to see the full prefix and the question.

Online paths must drop tokens before the question exists. Table 4 reports a **query-hold Pareto**: larger mid-stream hold improves recovery after query-aware tighten; final budget 1024 consistently beats 512 at

Table 3: LongBench slice (60 items @4k truncate): posthoc upper bound vs full. Peak for posthoc is full L (not a free systems win).

Arm	Mean F1	Substr hit	Peak	F1 / full
full	0.282	25%	= L	1.00×
posthoc@512	0.267	27%	= L	0.95 ×
posthoc@1024	0.284	28%	= L	~1.01×
posthoc@2048	0.285	27%	= L	~1.01×

Table 4: LongBench slice: online quality vs peak cache (query-hold Pareto).

Arm	Mean F1	Hit	Mean peak	F1 / full
novelty@512	0.179	10%	1024	0.64×
query_hold h1024→f512	0.199	18%	1536	0.70×
query_hold h1024→f1024	0.231	22%	1536	0.82×
query_hold h2048→f512	0.212	17%	~2531	0.75×
query_hold h2048→f1024	0.260	22%	~2531	0.92 ×
query_hold h4096→f1024	0.249	28%	~3876	0.88×

fixed hold. Best measured online point: hold 2048 → final 1024 ≈ **0.92**× full F1 at mean peak ~2.5k tokens—between novelty@1k (0.64×) and full@ L .

Interpretation. Open long-doc QA on this slice is not primarily “seed_valley cannot find answers when query-aware.” It is an **online retention** problem. Systems claims should be stated as a peak/quality curve, not as free near-full quality at peak~1k outside retrieval suites.

4.5 Systems resources (peak VRAM / latency)

Mid-depth needle, seed 0. Novelty stream@512 keeps peak VRAM and decode KV essentially flat:

Table 5: Peak resources: full chunked prefill vs sticky novelty stream@512.

L	full VRAM	nov. VRAM	full KV	nov. KV	nov. prefill
4k	8.5 GB	8.3 GB	569 MB	72 MB	1.6 s
16k	10.6 GB	8.3 GB	2296 MB	72 MB	5.9 s
40k	14.5 GB	8.3 GB	5754 MB	72 MB	17.6 s

Peak cache under novelty stays ~1024 tokens at all three lengths.

5 Adaptive policy (systems)

prefill_auto applies L -based budgets + per-model floors. With default discovery="novelty", single-needle stream@**512** is the measured multi-seed operating point through **40k**. Attn-only discovery still needs larger floors (~1536 multi-seed @4k). Hybrid Gemma: compress **full** layers only; sliding layers keep absolute prefix bookkeeping for score-pass.

6 Limitations and what we do *not* claim

6.1 What we claim

- On this **controlled multi-seed retrieval suite**, $\text{critical} \pm R^*$ is necessary/sufficient for $\varepsilon \approx 0$ single-fact recall.
- Stream failures at moderate budget are primarily a **query-unknown discovery** problem (oracle-online upper bound).
- Sticky surface novelty **closes most of that gap** through **40k** multi-seed at peak cache $\sim 1\text{k}$ on the primary model, with transfer smoke to other small instruct models including hybrid Gemma-4.
- On public LongBench, **posthoc query-aware compression** $\approx$ **full** at final $B=512$ (peak= L), while online novelty lags; **query_hold** recovers a measured Pareto (best $\sim 0.92 \times$ full F1 at peak $\sim 2.5\text{k}$).

6.2 What we do *not* claim

Table 6: Scope boundaries (honest).

Not claimed	Why
General long-context SOTA	No full RULER / LongBench / InfiniteBench leaderboard evaluation.
All task types	Primary suite is retrieval / needle-class (+ multi-needle, hops, prose, multidoc stress). Full reasoning chains, code repos, and full LongBench/RULER leaderboards untested.
Oracle-tight online budgets for free	Sticky novelty $\approx$ oracle quality on this suite at $@512$, but multi-secret packing ($\sim 80\%$ multi3) and suite-alignment remain.
Production memory stack	Fake-int8 is logical accounting only; no fused kernels or serving productization.
Novelty as universal “importance”	Exploits surface distinctness vs repetitive filler; ID-like filler / filler-like secrets can still confuse discovery.
Large models / training-free SOTA	Primary evidence is $\sim 3\text{--}4\text{B}$ instruct models on one GPU class.
Statistical finality	Multi-seed N modest ($5 \times 3 @4\text{k}$; $3 \times 3 \text{ long-}L$). Lab-grade, not a large meta-analysis.
$\varepsilon = 0$ beyond measured envelope	Sticky multi-seed measured through 40k ; longer L is extrapolation.
Peak VRAM = final decode KV	Streaming caps peak cache tokens ; weights and activations still dominate total VRAM.

6.3 Threats to validity

- **Greedy decode** and chat templates may interact with exact-match scoring.
- **Filler distribution** is synthetic; natural corpora may change novelty baselines.
- **Hybrid models**: only full layers are compressed; sliding-window behavior is infrastructure-sensitive.
- **Selection bias toward positive arms**: we iterated methods until discovery worked; negative results (scored capsules alone, non-sticky long- L) are reported but the path is research-in-the-loop.

7 Conclusion

Near-lossless training-free KV compression on retrieval tasks is structured by **critical local neighborhoods**. When the question is known, a simple attention valley scorer approximates the oracle keep set with small tax—and on a public LongBench slice, posthoc query-aware compression reaches $\sim$ full

quality at final $B=512$. When the question is **not** known—as in online streaming—the binding constraint is **discovery**/retention before the query. Sticky surface novelty is a lightweight query-unknown detector that, on our multi-seed suite, restores high success through **40k** context at a **flat $\sim 1\text{k}$ -token peak cache**. On open long-doc QA, quality becomes a **Pareto** in peak cache (query-hold $\sim 0.92\times$ full F1 at $\sim 2.5\text{k}$ peak). Honest next steps are better mid-stream pins and multi-entity packing—not larger single-needle grids alone.

Reproducibility

```
# Multi-seed H1 + scorer tax
python experiments/bench_paper_rigor.py --seeds 0,1,2,3,4 --ctx 4096

# Discovery gap / novelty
python experiments/bench_capsules.py --novelty --oracle-online \
  --skip-capsules --budgets 512
python experiments/bench_novelty_stress.py
python experiments/bench_novelty_longL.py \
  --ctx 16384,24576,32768,40960 --arms novelty --budgets 512

# LongBench posthoc UB + query_hold Pareto (60 items)
python experiments/bench_external_slice.py --source longbench --n 60 \
  --arms full,posthoc --budgets 512,1024,2048
python experiments/bench_external_slice.py --source longbench --n 60 \
  --arms full,novelty,query_hold --budgets 512 \
  --hold-budgets 1024,2048,4096 --final-budgets 512,1024

# Figure + transfer
python experiments/plot_paper_figures.py
python experiments/bench_transfer.py --model google/gemma-4-E4B-it
```

Primary model: Qwen/Qwen3-4B-Instruct-2507, transformers, CUDA bf16, RTX 3090.

Artifact index: results/FINDINGS.md.

Markdown twin: papers/PAPER_DRAFT.md.

Build PDF: from papers/, run powershell -File build_pdf.ps1.

References

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